



Lakshya JEE 2.0 (2024)

Limit, Continuity and Differentiability

DPP-04

- For $x \in \mathbb{R}$, $\lim_{x \rightarrow \infty} \left(\frac{x-3}{x+2} \right)^x$ is equals to:
 - e
 - e^{-1}
 - e^{-5}
 - e^5
- $\lim_{x \rightarrow 0} \left(\frac{4^x + 9^x}{2} \right)^{\frac{1}{x}}$ is equals to:
 - 2
 - 6
 - 16
 - 112
- Let $P = \lim_{x \rightarrow 0^+} (1 + \tan^2 \sqrt{x})^{\frac{1}{2x}}$, then $\log P$ is equal to:
 - $\frac{1}{2}$
 - $\frac{1}{4}$
 - 2
 - 0
- $\lim_{x \rightarrow 0} \left(\frac{1 + \tan x}{1 + \sin x} \right)^{\operatorname{cosec} x}$ is equal to
 - e
 - $\frac{1}{e}$
 - 1
 - None of these
- If $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2} \right)^{2x} = e^2$, then the value of a and b are:
 - $a = 1$ and $b = 2$
 - $a = 1, b \in \mathbb{R}$
 - $a \in \mathbb{R}$ and $b = 2$
 - $a \in \mathbb{R}, b \in \mathbb{R}$
- $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 2x - 1}{2x^2 - 3x - 2} \right)^{\frac{2x+1}{2x-1}}$ is equal to
 - 0
 - ∞
 - 1/2
 - None of these
- $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} \right)^x$
 - e^4
 - e^2
 - e^3
 - 1
- $\lim_{n \rightarrow \infty} (4^n + 5^n)^{1/n}$ is equal to
 - 4
 - 5
 - e
 - None of these
- $\lim_{x \rightarrow 0} \{(1+x)^{2/x}\}$ is equal to, (where $\{.\}$ denotes fractional part function)
 - $e^2 - 7$
 - $e^2 - 8$
 - $e^2 - 6$
 - None of these
- $\lim_{x \rightarrow 1} \left[\operatorname{cosec} \frac{\pi x}{2} \right]^{\frac{1}{1-x}}$ is equal to, (where $[.]$ is G.I.F.)
 - 0
 - 1
 - ∞
 - Does not exist
- $\lim_{m \rightarrow \infty} \left(\cos \frac{x}{m} \right)^m$ is equal to
 - 1
 - e
 - e^{-1}
 - None of these
- The value of $\lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos x}{x^4}$ is
 - 1/5
 - 1/6
 - 1/4
 - 1/2
- $\lim_{x \rightarrow 1} (2-x)^{\tan\left(\frac{\pi x}{2}\right)}$ is equal to
 - $e^{-2/\pi}$
 - $e^{1/\pi}$
 - $e^{2/\pi}$
 - $e^{-1/\pi}$

14. If $\lim_{x \rightarrow 0} \frac{\sin(\sin x) - \sin x}{ax^3 + bx^5 + c} = -\frac{1}{12}$, then a equals

- (1) 0
- (2) 1
- (3) 2
- (4) -1

15. If $\lim_{x \rightarrow 0} (x^{-3} \sin 3x + ax^{-2} + b)$ exists and is equal to zero then

- (1) $a = -3, b = \frac{9}{2}$
- (2) $a = 3, b = \frac{9}{2}$
- (3) $a = -3, b = -\frac{9}{2}$
- (4) $a = 3, b = -\frac{9}{2}$



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (3)
2. (2)
3. (1)
4. (3)
5. (2)
6. (3)
7. (1)
8. (2)

9. (1)
10. (2)
11. (1)
12. (2)
13. (3)
14. (3)
15. (1)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8eid80if>